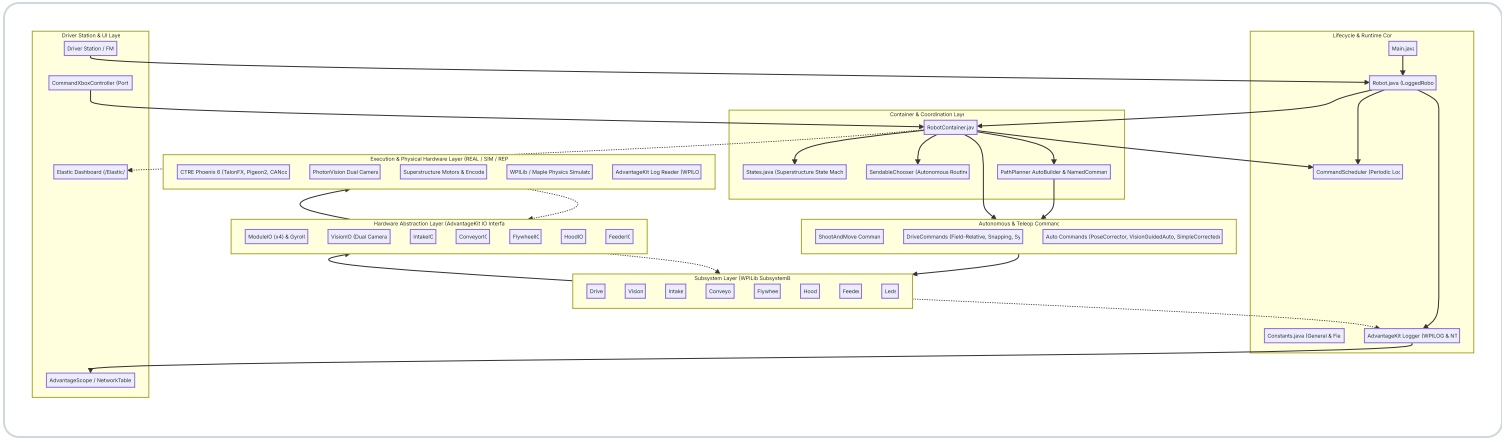
Robot-2026 Software Architecture Specification & Diagrams

Target Repository: `C:\Users\so2wi\Robotics\git\Robot-2026`
Team: FRC 3512
Core Frameworks: WPILib (Command-Based), AdvantageKit (Littleton Robotics), PathPlannerLib, CTRE Phoenix 6, PhotonVision, Elastic Dashboard

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1. High-Level System Architecture

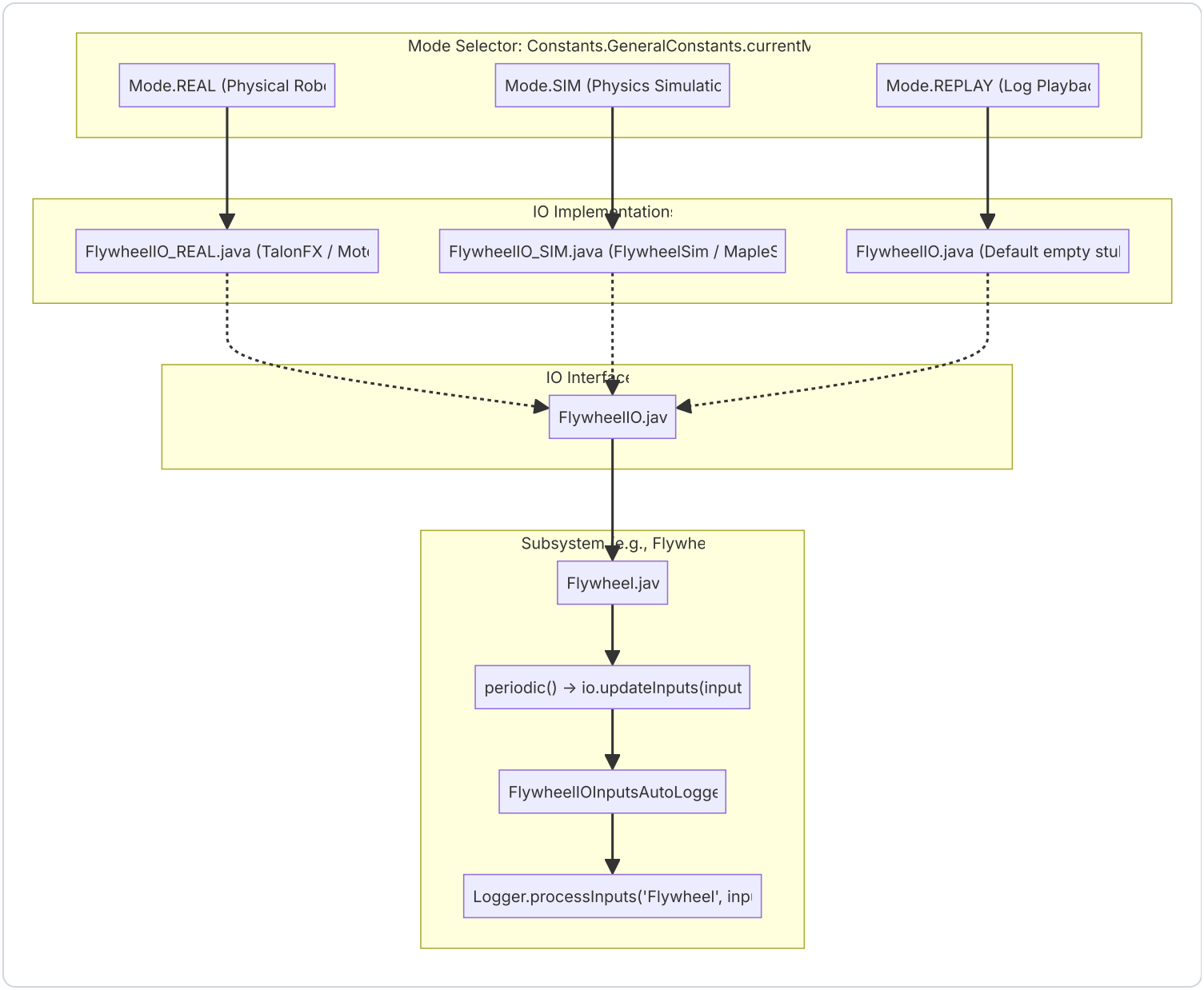
The robot software architecture follows the **AdvantageKit IO-Separation Layered Architecture** with the WPILib Command-based framework. It isolates hardware input/output behind abstract IO interfaces to enable deterministic log replay, physics simulation, and multi-mode robot operation.



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2. AdvantageKit Hardware Abstraction & Tri-Mode Architecture

Every mechanism in the robot implements the AdvantageKit pattern. Subsystems never interact directly with vendor hardware APIs. Instead, they define an interface and an auto-logged inputs data carrier (`@AutoLog`).


Mode Selector: Constants.GeneralConstants.currentM
Mode.REAL (Physical Robo
Mode.SIM (Physics Simulatic
Mode.REPLAY (Log Playbac
IO Implementation
FlywheelIO_REAL.java (TalonFX / Mot
FlywheelIO_SIM.java (FlywheelSim / MapleS
FlywheelIO.java (Default empty stul
IO Interface
FlywheelIO.jav
Subsystem (e.g., Flywhe
Flywheel.jav
periodic() → io.updateInputs(input
FlywheelIOInputsAutoLogge
Logger.processInputs('Flywheel', inp

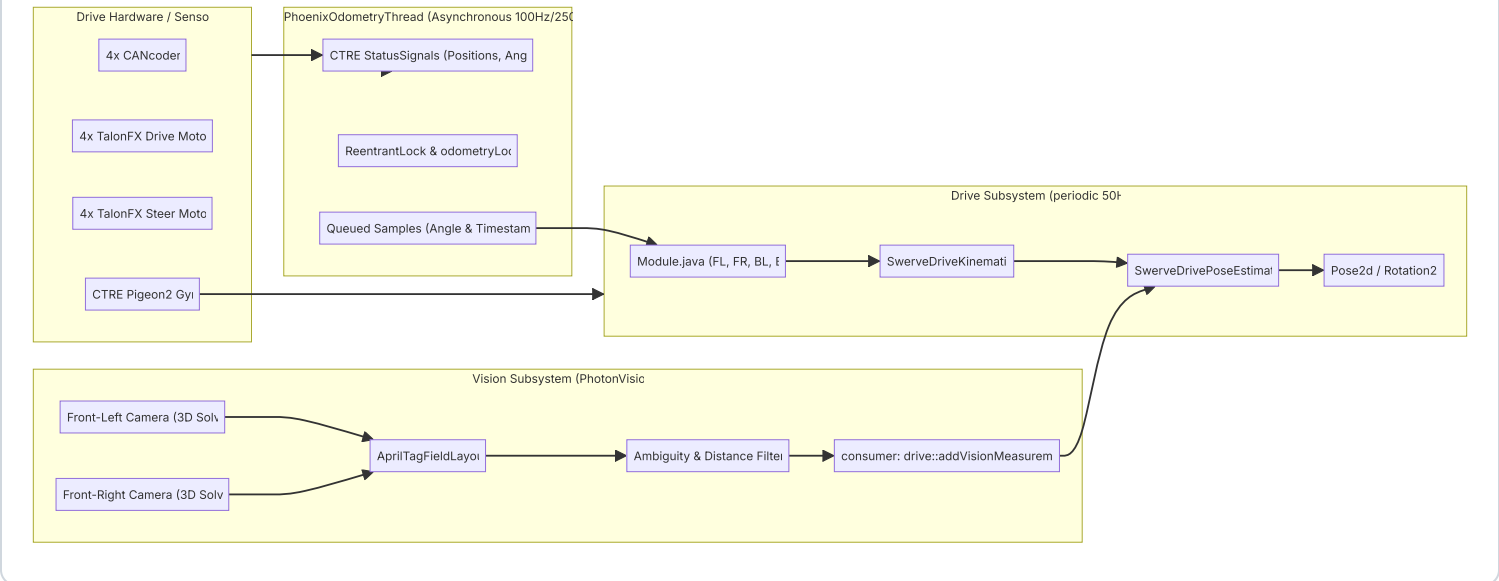
Subsystem IO Mapping Matrix

Subsystem	IO Interface	Real Hardware Implementation (`_REAL`)	Simulation Implementation (`_SIM`)
Drive (Swerve Modules)	`ModuleIO`	`ModuleIOTalonFX` (TalonFX + CANcoder)	`ModuleIOSim`
Drive (IMU)	`GyroIO`	`GyroIOPigeon2` (CTRE Pigeon 2)	GyroIO default stubs
Vision	`VisionIO`	`VisionIOPhotonVision` (FrontLeft, FrontRight)	`VisionIOPhotonVisionSim`
Intake	`IntakeIO`	`IntakeIO_REAL`	`IntakeIO_SIM`
Conveyor / Hopper	`ConveyorIO`	`ConveyorIO_REAL`	`ConveyorIO_SIM`
Shooter Drum (Flywheel)	`FlywheelIO`	`FlywheelIO_REAL`	`FlywheelIO_SIM`
Shooter Hood	`HoodIO`	`HoodIO_REAL`	`HoodIO_SIM`
Shooter Feeder	`FeederIO`	`FeederIO_REAL`	`FeederIO_SIM`

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3. Odometry, Vision Fusion & High-Frequency Thread Pipeline

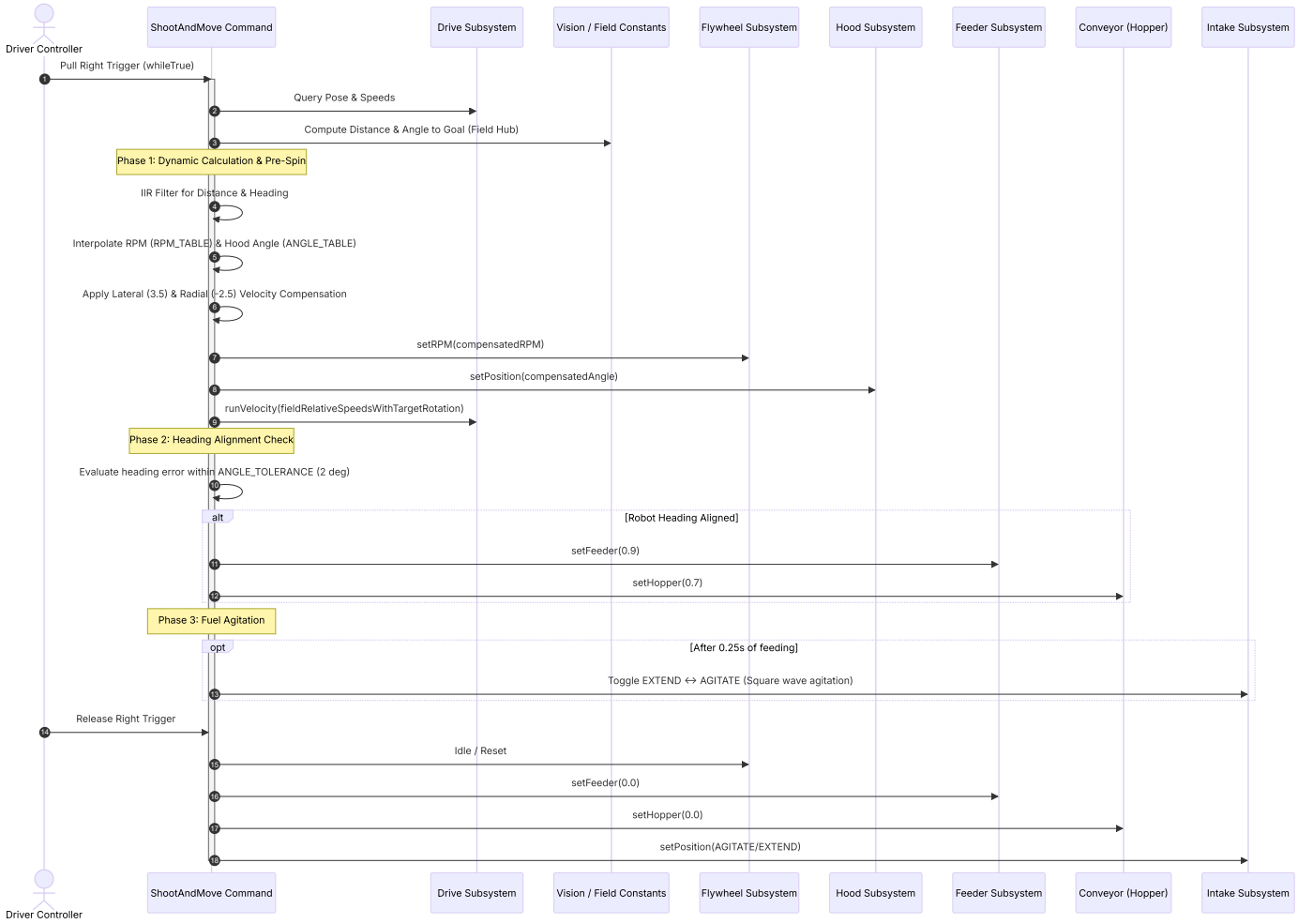
The drive subsystem features high-speed CANivore bus integration (250Hz) or RIO bus (100Hz) paired with a specialized odometry thread and dual PhotonVision AprilTag camera estimation.



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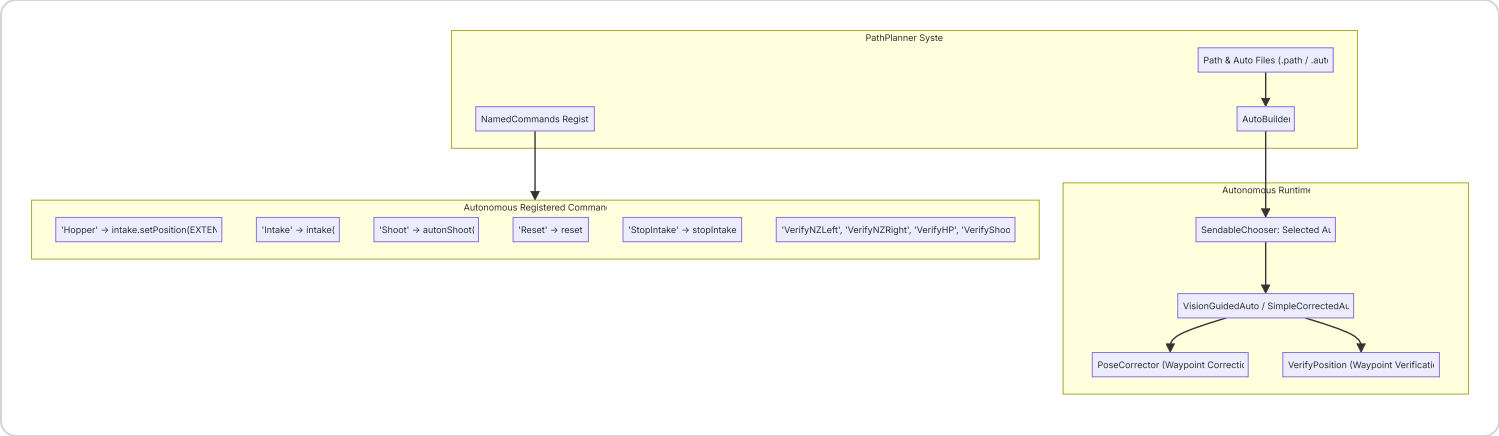
4. Superstructure Mechanism & `ShootAndMove` Pipeline

The robot features an integrated shooting pipeline (`ShootAndMove.java`) that allows automated targeting, flywheel acceleration, and continuous movement.



5. Autonomous Architecture & Vision Guidance

PathPlanner integration provides both standard predefined trajectory execution and dynamic vision-assisted correction.



6. Project Directory Map

- **Lifecycle**:
  - `Main.java` - Application bootstrap.
  - `Robot.java` - `LoggedRobot` handling WPILib loops, AdvantageKit logger setup, and scheduler execution.
  - `RobotContainer.java` - Dependency injection root, joystick bindings, auto registrations.
  - `Constants.java` - Field locations, target goal positions, and execution modes (`REAL` , `SIM` , `REPLAY`).
- **Subsystems**:
  - `subsystems/drive/` - 4-module swerve drive, high-frequency Phoenix odometry thread, kinematics, and SysId characterization.
  - `subsystems/vision/` - PhotonVision dual-camera AprilTag pose estimators and target tracking.
  - `subsystems/intake/` - Roller intake with motorized extension/retraction and ball agitation.
  - `subsystems/conveyor/` - Hopper conveyor routing game pieces toward the shooter.
  - `subsystems/shooter/drum/` - Dual-motor flywheel velocity regulation.
  - `subsystems/shooter/hood/` - Motorized adjustable launch angle hood.
  - `subsystems/shooter/feeder/` - Indexer/feeder wheel pushing balls into the flywheel.
  - `subsystems/led/` - Addressable LED status and animation system.
- **Commands**:
  - `commands/teleop/` - Field-relative joystick driving, hub aiming, and on-the-fly compensation in `ShootAndMove` .
  - `commands/auto/` - Waypoint pose correction (`PoseCorrector`), verification (`VerifyPosition`), and path wrappers (`VisionGuidedAuto`).
- **Utilities**:
  - `Elastic.java` - Rich notifications and tab switching for Elastic Dashboard.
  - `util/LocalADStarAK.java` - PathPlanner AD* pathfinder with AdvantageKit logging support.
  - `util/PhoenixUtil.java` - CTRE Phoenix error handling helper.